



DEPARTMENT OF COMPUTER SYSTEMS ENGINEERING

a. Personal Information			
Name: Prof. Dr. Shahnawaz Talpur			
Father's Name: Mir Abdul Sattar Talpur			
1.	i. Field of Specialization: Computer Software & Theory ii. Research Interests: High Performance Computing		
2.	Designation with Grade and department / institute: Professor (BPS-21), Department of Computer Systems Engineering, Mehran University of Engineering & Technology, Jamshoro.	3.	Administrative Post (Additional) Co-Director Institute of Information and Communication Technology MUET.
4.	Domicile: Hyderabad	5.	Date of Birth: 05.11.1977
6.	Religion: Islam	7.	Gender: Male
8.	PEC / PCATP Registration No.: COMP/1545	9.	Passport No.: AD 7730624
10.	Date of Joining of University: 04-01-2006	11.	Status of PhD Supervision: (HEC Approved / HEC Registered)
12.	Tel.: Off.: 022-2772250-60 (4201 / 4203 / 6202) Fax: Res.: 022-2106875 Cell: 0321-3008200	13.	Email Address: Official: shahnawaz.talpur@faculty.muett.edu.pk Personal: mirshan35@hotmail.com
14.	Marital Status: Married	15.	CNIC No.: 41303-5808062-3
16. HEC ENGAGEMENTS Member Accreditation/Inspection committee NCEAC Program Evaluator ENGAGEMENT IN OTHER UNIVERSITIES Member Board of Studies BAHAUDDIN ZAKARIYA university Multan Member Board of Studies QAID AWAM university Nawabshah Member Board of Studies SUFI university Bhitt Shah Evaluator PHD Thesis Computer Science / Information Technology NED university Karachi			
Research / Project Students Under supervision			
Undergraduate	Masters	PhD	
Several groups	20(completed) 08 (In progress)	03 (Completed) 1. Title: Scratchpad Memory Management Mechanism for Reconfigurable Cache/Spm Multi-Core Processors. 2. Learning from Peer Mistakes: Collaborative UML-Based Intelligent Tutoring System 3. Clustering Uncertain Overlapping Symptoms of Multiple Disease in Clinical Diagnosis 4. Design and Development of Privacy	

		Preserving Framework for Detection and Prevention of Cyber threats in User-Centric Data Models (In Progress)
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b. Education (Please start from Graduation Degree onwards)				
Date		Name & Address of Institution	Academic Degree / Diploma / Certificate Obtained	Subject / Field of Study
From	To			
1996	Year 2001	Mehran University of Engineering & Technology, Jamshoro	Bachelor of Engineering	Computer Systems
2003	Year 2005	Mehran University of Engineering & Technology, Jamshoro	Master of Engineering	Information Technology
2010	2014	Beijing Institute of Technology, China	PhD	Computer Science

c(i). Academic Experience (Please start with the most recent Experience)				
Designation / Title	Date		Name of Organization	Brief Job Description
	From	To		
HOD (Computer Engineering) /	2019	2026	Department of Computer Systems Engineering, MUET, Jamsho	Managing the academic, administrative, and strategic activities ensure Quality education
Professor	Nov 2024	Till Date	Department of Computer Systems Engineering, MUET, Jamshoro	Research & Teaching
Associate Professor	Oct 2016	Nov 2024	Department of Computer Systems Engineering, MUET, Jamshoro	Research & Teaching
Assistant Professor	Oct 2009	Oct 2016	Department of Computer Systems Engineering, MUET, Jamshoro.	Research & Teaching
Lecturer	Jan. 2006	Oct. 2009	Department of Computer Systems Engineering, MUET, Jamshoro.	Teaching
IT Manager (I.T Incharge)	June 2005	Jan. 2006	Pinnacle Distilleries Pvt	Managing IT Dept, Network Administration
Franchise Manager	Jan,2005	June 2005	Warid Telecom Franchise	Administration and Customer care
Systems Engineer	Nov, 2003	Jan, 2005	Nordtec International	System Trouble shooting
I.T Incharge	Oct, 2001	Oct, 2003	Ali Associates	IT manager & Database Administration

c(ii). Administrative Experience (Please start with the most recent Experience)				
Designation / Title	Date		Name of Organization	Brief Job Description
	From	To		

Chairman Computer Systems Engineering Department	2019	Till Date	Mehran UET, Jamshoro	Oversee and lead the Computer Systems Department as Chairman, managing a team of 20 staff members and 35 faculty while guiding the successful delivery of two bachelor's programs, Computer Systems Engineering and Computer Science, with a total enrollment of 500 students.
Co-Director IICT Additional Charge	2018	Till Date	Mehran UET, Jamshoro	Overseeing and directing administrative operations and academic activities for six postgraduate programs in computer-related fields, including curriculum management, seminar organization, examination coordination, and supervision of office staff.
Adjunct Professor	2022	2022		Involved in research with PhD students
IT Manager (I.T Incharge)	June 2005	Jan. 2006	Pinnacle Distilleries Pvt, Matli, Dist. Badin	Managing IT Dept, Network Administration
Franchise Manager	Jan,2005	June 2005	Warid Telecom Franchise, Gul Center Hyderabad	Operational and office management

c(iii). Industrial / Field Experience (Please start with the most recent Experience)				
Designation / Title	Date		Name of Organization	Brief Job Description
	From	To		
IT Manager (I.T Incharge)	June 2005	Jan. 2006	Pinnacle Distilleries Pvt, Matli, Dist. Badin	Intranet and Extranet Implementations of Computer network between the sites area and Head office

Table No. 1

Details of Research Papers Published in Journals:

S: NO	ISSN	PAPER TITLE	IMPACT FACTOR	HEC RECOGNIZED	JOURNAL/ CONFERENCE
1.	0368-492X	Impact of blockchain on deception of source information	2.4	Yes	Journal
2.	2376-5992	Clustering uncertain overlapping symptoms of multiple diseases in clinical diagnosis	3.8	Yes	Journal
3.	0368-492X	Impact of blockchain on deception of source information	2.4	Yes	Journal
4.	0141-9331	An interactive and dynamic scratchpad memory management strategy for multi-core processors.	1.9	Yes	Journal
5.	2376-5992	Multi-step rainfall forecasting using deep learning approach	3.8	Yes	Journal
6.	2413-7219	Management of scratchpad memory using programming techniques	0.6	Yes	Journal
7.	0141-9331	State of art innovative technique for management of scratchpad memory (scratch).	1.9	Yes	Journal
8.	2254-4143	An automated system for traffic sign recognition using convolutional neural network.		Yes	Journal

9.	0967-0912	Exploring grouped coherence for clustered hierarchical cache. In The Journal of Supercomputing, Springer.	3.3	Yes	Journal
10.	2413-7219	An intelligent service-based layered architecture for eLearning and assessment	0.6	Yes	Journal
11.	2413-7219	Mining frequent item sets in asynchronous transactional data streams over time sensitive sliding windows model	0.6	Yes	Journal
12.	2413-7219	Design and co-simulation of depth estimation using simulink HDL coder and modelsim.	0.6	Yes	Journal
13.	1573-7543	Achieving self-aware parallelism in stream programs.	3.6	Yes	Journal
14.	0142-0615	Evaluation of solar energy resources by establishing empirical models for diffuse solar radiation on tilted surface and analysis for optimum tilt angle for a prospective location in southern region of Sindh, Pakistan.	5.0	Yes	Journal
15.	1573-0484	Exploiting controlled-grained parallelism in message-driven stream programs.	2.5	Yes	Journal
16.	2502-4760	Input-Current and Load Voltage Sharing in Input-Parallel Output-Series Connected Boost Half Bridge DC-DC Converter Using Stable Control Scheme	0.247	Yes	Journal
17.	2502-4760	Intra-Inter Triplet Object Interaction Mechanism in Triplet-Based Hierarchical Interconnection Network	0.247	Yes	Journal
18.	2502-4760	Research on Dependable Ionizing Radiation Protection based on Model	0.247	Yes	Journal
19.	2502-4760	Instruction Pipeline Efficient Mechanism with maximum hit ratio	0.247	Yes	Journal
20.	2502-4760	A Long-wire-connected and Multi-channel 3D Network-on-chip Design for Many-core System	0.247	Yes	Journal
21.	1881-803X	Modeling And Resolving Lock Contention for Multi-Threaded Systems	0.261	Yes	Journal
22.	2523-0379	Segmentation and Classification of Calcification and Hemorrhage in the Brain using Fuzzy C-Mean and Adaptive Neuro- Fuzzy Inference System		Yes	Journal
23.	2523-0379	Future Infrastructure of IOT: Large Scale Deployments of IOT Components		Yes	Journal
24.	1013-5316	Nonlinear Transient Characterization of Third Order Voltage Switched Cp-Pll		Yes	Journal
25.	2791-0547	Redundant Capability Analysis of input-series output-series Multi-modular DC-DC converter		Yes	Journal
26.	1605-8607	Double Channel CNN Based Tomato Plant Leaf Disease Detection.		Yes	Journal
27.	2618-1630	Review of Peer Feedback in Collaborative Tutoring Systems		Yes	Journal
28.	2618-1630	AI-Based Predictive Tool-Life Computation in Manufacturing Industry		Yes	Journal
29.	2618-1630	Identification of Real and Fake Reviews Written in Roman Urdu.		Yes	Journal
30.	2618-1630	Classification of Amputee EMG Signals Using Machine Learning Techniques		Yes	Journal
31.	2618-1630	Utilizing Machine Learning for Detecting Cyber Bullying in Social Media		Yes	Journal
32.	2456-3307	A Discriminative Image Registration Model with Hybrid Feature Detector.		Yes	Journal
33.	2709-6130	Detection & Quantification of Lung Nodules Using 3D CT Images.		Yes	Journal
34.	2618-1630	Non-invasive EEG based Feature Extraction framework for Major Depressive Disorder analysis.		Yes	Journal
35.	2064-9622	Lightweight multi-agent framework for a cluster-based wireless sensor network.		Yes	Journal

36.	1605-8607	Coronavirus Detection using Deep Pre-trained Model on Chest X-ray Imaging		Yes	Journal
37.	2456-3307	Comparative Analysis of LSTM, BiLSTM and ARIMA for Time Series Forecasting on 116 years of Temperature and Rainfall Data from Pakistan.		Yes	Journal
38.	2321-1857	A Comparison of Proximity and Feature Based Segmentation of 3D MRI Images for Brain Tumor Detection		Yes	Journal
39.	2520-739	Early Detection of Type 2 Diabetes using supervised machine learning.		Yes	Journal
40.	2411-0930	Critical Analysis Of 3D Manycore Memory Architecture Using Mapreduce		Yes	Journal
41.	2581-3048	A Hybrid Architecture Based on Deep Learning for Object Recognition for Autonomous Driving.		Yes	Journal
42.	0975 – 8887	A Qualified review of ML and DL Algorithms for Bearing Fault Diagnosis		Yes	Journal
43.	0975 – 8887	Predicting the Best Team Players of Pakistan Super League using Machine Learning Algorithms		Yes	Journal
44.	2521-5027	Computer Assisted Detection of Elbow Deformation in Digital X-rays		Yes	Journal
45.	2414-1399	Shape Based Image Retrieval Using Fused Feature		Yes	Journal
46.	0976-2191	Forged character detection datasets: passports, driving licences and visa stickers.		Yes	Journal
47.	2521-5027	Spare Deep Learning and Hyper Spectral Image Classification		Yes	Journal
48.	2582-5208	Deep Q-Learning Based Automated Parking Assistance Model for Four-Wheel Light Transport Vehicle		Yes	Journal
49.	2618-1630	Automated Objects Delivery System for Interior Locale using Line Following Robot with Optimized Security Parameters		Yes	Journal
50.	1813-1743	Redundant Capability Analysis of input-series output-series Multi-modular DC-DC converter		Yes	Journal

Table No. 3

Details of Research Papers Presented at Conferences

S: NO	ISBN	PAPER TITLE	HEC RECOGNIZED	JOURNAL/ CONFERENCE
1	979-8-3503-3846-1	Myocardial D-Spect Image Segmentation Using RSF Model and Kernal Difference Operator.	Yes	Conference
2	978-1-6654	Speech Emotion Recognition Using Deep Learning Hybrid Models.	Yes	Conference
3	978-1-6654-8939-3	An MFCC-based secure framework for voice assistant systems.	Yes	Conference
4	978-1-6654-7105-3	Twitter Mining based Forecasting of cryptocurrency using sentimental analysis of Tweets	Yes	Conference
5	978-1-6654-7105-3	Analysis of Dynamic Host Control Protocol Implementation to Assess DoS Attacks.	Yes	Conference
6	978-969-23372-2-9	Frequency Based Approach for Railway Track Faults Monitoring in Pakistan	Yes	Conference

7	978-969-23372-2-9	Automatic Detection of Humerus Deformation in X-ray Imagery.	Yes	Conference
8	978-969-23372	Analysis of Usability Issues of Q-OBE: Teachers' perspective	Yes	Conference
9	978-1-6654-2386-1	Machine Learning Approach for Classification of DHCP DoS Attacks in NIDS.	Yes	Conference
10	978-981-15-3284-9	Blockchains technology analysis: applications, current trends and future directions—an overview	Yes	Conference
11	978-1-7281-4970	Detecting Faulty Sensors by Analyzing the Uncertain Data Using Probabilistic Database	Yes	Conference
12	978-1-7281-7283-5	Short-term prediction model for multi-currency exchange using artificial neural network	Yes	Conference
13	978-1-7281-4082-7	Reconstruction and archival approaches with 3D visualization of Cultural Heritage in Museums of Sindh Region	Yes	Conference
14	978-3-319-77712-2	A Novel Curvature Feature Embedded Level Set Method for Image Segmentation of Coronary Angiograms.	Yes	Conference
15	978-1-5386-1978	Evaluation and analysis of Network Coding at Network Layer.	Yes	Conference
16	978-1-5386-4844-5	Brain tumor segmentation using 3d magnetic resonance imaging scans	Yes	Conference
17	2473-3520	Analysis of Performance of Instruction Pipeline with Transactional Slice Mechanism in CMP	Yes	Conference
18	2166-0670	Improving router efficiency in network on chip triplet-based hierarchical interconnection network with shared buffer design.	Yes	Conference
19	978-1-4799-4922-9	Optimal Pipeline Performance via Transactional Slice with No Branch Prediction Overhead	Yes	Conference
20	978-3-642-35605-6	Communication Locality Analysis of Triplet-Based Hierarchical Interconnection Network in Chip Multiprocessor.	Yes	Conference
21	978-1-4673-2164-8	Express router microarchitecture for triplet-based hierarchical interconnection network.	Yes	Conference
22	979-8-3503-5236-8	Comparative Analysis of Deep Learning Architectures for Blood Cancer Classification	Yes	Conference

Table No. 4

Detail Books and Chapter Published

S.No.	ISBN Number	Title	Publisher Name	Index
1	978-1-80592-062-5	Explainable AI (XAI) in Air Traffic Monitoring Systems	emarland	Web of Science, Scopus, and Ei Compendex
2	9781668477021	AI and Its Convergence with Communication Technologies	IGI Global Scientific Publishing bookstore.	Web of Science, Scopus, and Ei Compendex.

Table No. 5

Details of Conferences / Seminars / Symposia / Workshops attended in Pakistan:

Sr. No.	Name of Event	Type of Event	Organizing Agency	Venue	Date(s) of Event
1.	The entrepreneur's Path: Your Gateway to Business Excellence	Workshop	MUSTID	MUSTID	18-22 DEC 2023
2.	OBE Training	Training	IICT	MUET	1 OCT2022
3.	PhD supervisor Capacity Building Program	Workshop	HEC, Pakistan	HEC Regional	4-6 March 2019
4.	IBM Rational Quality Suite	Training	Avanza	MUET	22-30 April 2010
5.	Workshop on Academic & Technical Writing Skills	Workshop	AL-BARR Academy	CS & SW dept. MUET	2 nd May 2009
6.	Creating Social Semantics Web 2.0 Information Space	Workshop	IMTIC'08	MUET	12 th April 2008
7.	Wireless Sensor Networks: From Technology to Reality	Workshop	IMTIC'08	MUET	13 th April 2008
8.	Fiber Optics	Training	ComBit telecom	MUET	08-11 May 2008
9.	Staff Development Course	Workshop/Training	National Academy HEC	MUET	26 March-26 April 2007
10.	HEC-Digital Library Program	Workshop	HEC/MUET	MUET	12 th November 2007
11.	Advance Communication and study skills	Pilot Course	HEC	MUET	13-18 November 2006
12.	"21" All Pakistan Electrical & Electronics Engineers Students' Seminar	Seminar	Institute of Electrical & Electronics Engg.	MUET	21 st January

Table No. 6

Details of Conferences / Seminars / Symposia / Workshops attended outside Pakistan:

Sr. No.	Name of Event	Type of Event	Organizing Agency	Venue and Country	Date(s) of Event
1.	PLDI + ECOOP 2012 Conference	Conference	ACM SIGPLAN	Beijing, China	11-16, June 2012
2.	Context-Oriented Programming COP'12	Workshop	ACM SIGPLAN	Beijing, China	11 th June 2012
3.	LaME	Workshop	ACM SIGPLAN	Beijing, China	13 th June 2012
4.	Evaluate	Workshop	ACM SIGPLAN	Beijing, China	15 th June 2012
5.	Benchmarking and Performance Evaluation	Workshop	ACM SIGPLAN	Beijing, China	16 th June 2012

6.	The 9th IFIP International Conference on Network and Parallel Computing	Conference	Seoul National University of Technology, Korea Mokwon University, Korea	Gwangju, Korea	06-08 Sept. 2012
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Table No. 8

Details of International Proceedings:

Sr. No.	Title of Research Journal	Name and Address of Publisher	Year
1.	Network and Parallel Computing	LNCS Springer	2012

Table No. 9

Details of National Research / Collaborative Research Grants / Projects Carried Out:

Sr. No.	Name of Research / Collaborative Research Grant / Project	Name of Funding / Collaborating Agency	Organizing Agency	Total Amount of Grant / Project	Date of Commencement	Duration
1.	N/A					

Table No. 10

Details of International Research / Collaborative Research Grants / Projects Carried Out:

Sr. No.	Name of Research / Collaborative Research Grant / Project	Name of Funding / Collaborating Agency & Country	Organizing Agency	Total Amount of Grant / Project	Date of Commencement	Duration
1.	N/A					

Table No. 11

Details of PHD / ME / MS / M.Phil. / Groups or Projects of Undergraduate Students Supervising & Supervised:

Students Status	PhD Students		ME / MS Students		M.Phil. Students		Undergraduate Students			Total	
	M	F	M	F	M	F	No. Groups / Projects	M	F	M	F
Students Supervising											
Students Supervised and Awarded Degrees	2	1	13	11				64	63	79	75

Table No. 12

Details of International Awards / Medals / Distinctions Received:

Sr. No.	Name of Faculty Member	Name of Award / Medal / Distinction	Name of Authority & Country Giving Award / Medal / Distinction	Date of Award / Medal / Distinction
1.	Talpur	CSC	Chinese Govt. Scholarship	2010

Table No. 13

Details of Memberships of National Professional Bodies / Councils / Committees / Organizations:

Sr. No.	Name of Bodies / Councils / Committees / Organizations	Date and Duration of Membership
1.	Pakistan Engineering Council	2001-Life time

Table No. 14

Details of Memberships of International Professional Bodies / Councils / Committees / Organizations:

Sr. No.	Name of Bodies / Councils / Committees / Organizations and Country	Membership Status	Date and Duration of Membership
1.	International Association of Engineers (IAENG), Hong Kong	Member	Oct 2011